



# Algebraic estimation-based model-free control for variable-speed wind turbines

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## ABSTRACT

This research study validates the recently invented Model-free Control (MFC) approach via algebraic estimation to a variable-speed wind turbine. Instead of counting on adaption rules or neural networks, this novel technique employs an ultra-local model for rapid algebraic parameter estimation, requiring no statistical expertise about system models, disturbances, or interdependence between inputs and outputs. In low-speed winds, a model-free controller is designed based on a wind estimator to track the optimum value of power coefficients. In high-speed winds, MFC is extended to a multi-variable model-free controller (MV MFC) in pitch and torque to regulate the produced power around its nominal value. The approach's key advantage lies in eliminating complex non-linear models, ensuring stability, and reducing non-linearities, uncertainties, and perturbations. The controllers have been tested with a wind turbine mathematical model and validated with FAST simulator in the presence of measurement noise and disturbances. The results show the proposed controller's superiority over the existing techniques and improvement of power capture.

## 1. Introduction

In recent decades, renewable energies have gained attention in the energy market to meet the growing global demand for clean sources [1]. Wind energy is recognized as a widely accessible and economically viable power source globally. It serves as a practical option for addressing challenges related to the global energy crisis and climate change [2]. The challenges of harnessing power from wind energy are intriguing, and many countries globally have allocated funds to address these issues [3]. Ensuring the efficiency and durability of wind turbines necessitates the development of robust controllers for this large-scale equipment. High-efficiency techniques are imperative to extract the maximum power from the available wind, where, controllers must ensure the long life of these pricey turbines [4]. Variable-pitch and variable-speed wind turbines are particularly interesting in their ability to increase energy extraction while preventing turbine damage [5,6]. Consequently, an efficient control technique is required to optimize power generation while minimizing structural loads.

The earliest straightforward fixed-pitch and fixed-speed rotors operated without the need for control regulation. These machines could only operate efficiently at nominal wind speeds. For that, numerous other control strategies were developed for the next generation of them.

There are many controllers in the literature that are based on linear-time invariant (LTI) models. Over the past decade most research has emphasized the use of conventional PID controllers for linear system wind turbines [7]. More recent attention has focused on the provision of the Linear Quadratic (LQ) and Linear Quadratic Gaussian control techniques [8,9]. In another major study, robust controllers have been reported in [10,11]. Some studies have focused on developing classical control strategies. Model predictive control (MPC) is an advanced controller approach that has been used to address pitch-angle restrictions [12]. By drawing on the concept of fuzzy logic, Bououden has been able to show an effective nonlinear modeling technique that is particularly suitable for systems whose dynamics vary according to the operating point [13]. A fault tolerant controller is addressed to control rotor speed and power of a wind turbine in [14].

Fliess and Sira-Ramírez conducted a development of algebraic techniques that ensures a precise identification of the system's parameters, states, failures and input-output perturbations. The foundational principles of the method for examining linear and non-linear system cases were reported in [15]. The recently introduced intelligent PID (i-PID) controller stands as one such example, demonstrating its efficacy in various applications when compared to the conventional PID. Fliess

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| List of Main Acronyms and Parameters |                                                |
|--------------------------------------|------------------------------------------------|
| MFC                                  | Model-Free Control                             |
| WT                                   | Wind Turbine                                   |
| PID                                  | Proportional–Integral–Derivative               |
| LQG                                  | Linear Quadratic Gaussian                      |
| MPC                                  | Model Predictive Control                       |
| FAST                                 | Fatigue, Aerodynamics, Structures, Turbulence  |
| MFCE                                 | Model-Free Control based wind Estimator        |
| MV MFC                               | Multi-Variable Model-Free Control              |
| i-PID                                | Intelligent Proportional–Integral–Derivative   |
| ISC                                  | Indirect Speed Control                         |
| ATF                                  | Aerodynamic Torque Feedforward                 |
| NSSFE                                | Nonlinear Static State Feedback with Estimator |
| SNR                                  | Signal-to-Noise Ratio                          |
| FIR                                  | Finite Impulse Response                        |
| CART                                 | Controls Advanced Research Turbine             |
| NREL                                 | National Renewable Energy Laboratory           |
| DOF                                  | Degree of Freedom                              |
| $\omega_r$                           | Rotor speed                                    |
| $\omega_{r\text{opt}}$               | Optimal rotor speed                            |
| $T_g$                                | Generator torque                               |
| $\beta$                              | Blade pitch angle                              |
| $C_p$                                | Power coefficient                              |
| $C_q$                                | Torque coefficient                             |
| $\lambda$                            | Tip-speed ratio                                |
| $J_t$                                | Total inertia                                  |
| $\eta_{\text{elec}}$                 | Electrical efficiency                          |
| $\eta_{\text{aero}}$                 | Aerodynamic efficiency                         |
| $F(t)$                               | Unmodeled dynamics and disturbances            |
| $\beta_1, \beta_2$                   | Scaling parameters for MV MFC                  |
| $\lambda_{\text{opt}}$               | Optimal tip-speed ratio                        |
| $n_g$                                | Gearbox ratio                                  |
| $R$                                  | Rotor radius                                   |
| $P_a$                                | Aerodynamic power                              |
| $P_{a\text{opt}}$                    | Optimal aerodynamic power                      |
| $v$                                  | Wind speed                                     |

and Join carried out the model-free control (MFC) technique by extending beyond classical PID controllers the intelligent PID controllers (i-PID). These controllers accommodate the unknown dynamics of the system and prove beneficial for improving reference tracking [16]. This resolution is grounded in a novel formulation of the ultra-local model-based control approach. Given that MFC operates without the need for a nominal model, the control methodology inherently demonstrates robustness.

Numerous successful research applications have examined Model-Free Control (MFC) across diverse domains, including vehicle system control [17], magnetic levitation systems [18], quadrotor helicopters [19,20], fuel power generation [21], hybrid unmanned aerial vehicles [22], exoskeleton-upper limb systems [23], industrial electro-hydraulic systems [24], and autonomous underwater vehicles [25]. Recent studies further highlight MFC's growing relevance in advanced mobility and automation, such as extremum-seeking adaptive MFC for automated vehicle path tracking [26], data-driven yaw-rate control achieving superior robustness under varying loads [27], cruise control for ground vehicle systems under uncertainty [28], and model-free trajectory tracking for tail-sitter UAVs in hovering mode [29].

The aim of this paper is to introduce a recent and effective control strategy for variable-speed wind turbines based on model-free control (MFC). The motivation behind this approach lies in the increasing complexity and nonlinearity of modern wind turbine systems, which make traditional model-based control techniques difficult to implement effectively. Conventional controllers such as PID, LQG, or MPC depend

on accurate mathematical models and extensive parameter tuning. However, in wind energy applications, these models are often difficult to obtain or maintain due to unpredictable environmental conditions, structural flexibilities, and unmodeled dynamics [30].

Model-free control offers a compelling alternative by eliminating the need for detailed system modeling. It replaces the unknown system dynamics with an ultra-local model and uses real-time algebraic estimation to identify a single aggregated parameter. This makes the control design significantly simpler, more robust to disturbances, and easier to implement in real-world conditions. Motivated by the success of MFC in various industrial applications—such as robotics, vehicle control, and process systems—this work applies and extends the approach to the wind energy domain.

In the below-rated wind speed region, where the goal is to maximize energy capture, the proposed controller Model-free control based wind estimator (MFCE) regulates rotor speed via generator torque using estimated wind speed. In the above-rated region, where the objective is to limit power and reduce mechanical stress, the method is extended to a multivariable model-free controller (MV-MFC) that acts on both torque and pitch angle.

Furthermore, the model-free approach provides robustness against modeling uncertainties and measurement noise, and facilitates real-time implementation due to its low computational cost. These characteristics make it particularly suitable for wind turbines, where flexibility, adaptability, and simplicity are crucial. The proposed methods are validated using both a simplified mathematical model and the aeroelastic FAST simulator, demonstrating superior performance compared to classical controllers.

The rest of the paper is organized as follows. Section 2 illustrates a brief overview of the wind turbine model requirements, the simplified mathematical model, the aeroelastic FAST simulator, and the problem formulation. Section 3 examines an overview of the developed controllers, the estimation process of the wind, and the design criteria. In Section 4 the findings of the research are presented using the mathematical model, validated with FAST simulator, and compared with existing control strategies. Finally, The conclusion and perspectives are given in the Section 5.

## 2. Wind turbine system modeling

The control model is structured on the wind turbine dominating dynamics, which includes drive-train dynamics, generator dynamics, and rotor aerodynamics [31]. The following subsections investigate these dynamics:

### 2.1. Aerodynamic model

Rotor power, a measure of wind energy over time, hinges on factors like rotor speed ( $\omega_r$ ), blade pitch angle ( $\beta$ ), wind speed ( $v$ ), aerodynamic power ( $P_a$ ), and torque ( $T_a$ ). A non-linear function, as per [32], defines their interdependence:

$$P_a = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3 \quad (1)$$

where  $\rho$  represents the air density,  $R$  corresponds to the blade length (also known as the rotor radius), and  $\lambda$  signifies the tip speed ratio such that:

$$\lambda = \omega_r R / v \quad (2)$$

$C_p(\lambda, \beta)$  denotes the power coefficient, representing wind turbine efficiency via a non-linear equation. The aerodynamic torque coefficient relates to the power coefficient as follows:

$$T_a = P_a / \omega_r = \frac{1}{2} \rho \pi R^3 C_q(\lambda, \beta) v^2 \quad (3)$$

where,

$$C_q(\lambda, \beta) = \frac{C_p(\lambda, \beta)}{\lambda}$$

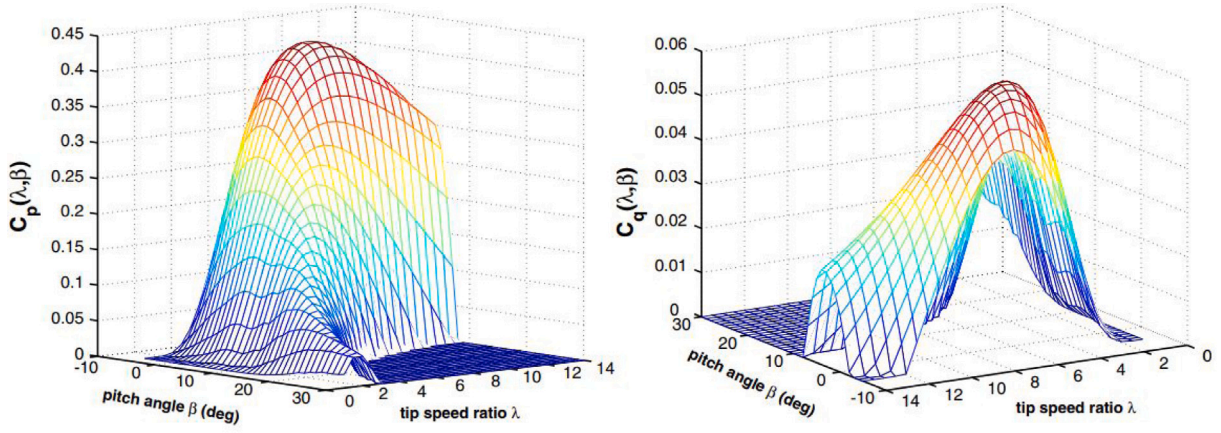


Fig. 1. Power and torque coefficients of CART wind turbine. ( $C_p(\lambda, \beta)$ ,  $C_q(\lambda, \beta)$  curves.)

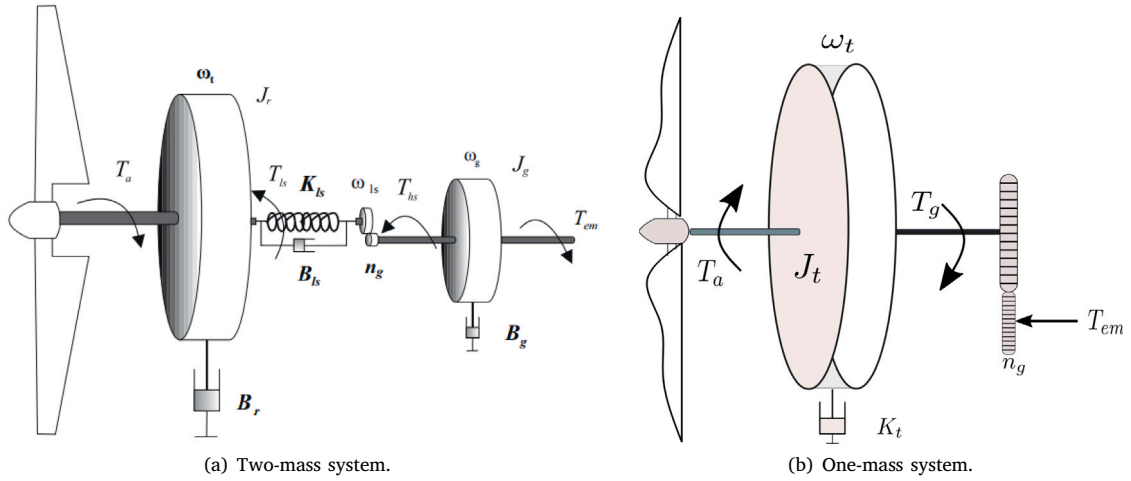


Fig. 2. Dynamics of the drive-train system.

The research study presents the torque and power coefficient surfaces for the analyzed wind turbine, as depicted in Fig. 1. These surfaces were generated through the utilization of blade-element moment theory, employing the code for wind turbine performance, WT-PERF. [33], which was developed by the National Renewable Energy Laboratory (NREL).

The exhibited behavior of the rotor, which is propelled by the aerodynamic torque  $T_a$  at a speed of  $\omega_r$ , demonstrates its dynamic response as follows: w

$$J_r \dot{\omega}_r = T_a - T_{ls} - K_r \omega_r \quad (4)$$

The torque exerted on the rotor by the low-speed shaft, represented as  $T_{ls}$ , functions as a braking torque (refer to Fig. 2(a)). This torque is generated as a result of torsional and frictional effects arising from the disparity between  $\omega_r$  and  $\omega_{ls}$ :

$$T_{ls} = B_{ls} (\theta_r - \theta_{ls}) + K_{ls} (\omega_r - \omega_{ls}) \quad (5)$$

The torque  $T_{hs}$  on the high-speed shaft drives the generator, whereas the electromagnetic torque  $T_{em}$  from the generator acts to oppose or decelerate its motion.

$$J_g \dot{\omega}_g = T_{hs} - K_g \omega_g - T_{em} \quad (6)$$

Given the assumption of an ideal gearbox, characterized by a transmission ratio denoted as  $n_g$ , the following relationship holds

$$n_g = \frac{T_{ls}}{T_{hs}} = \frac{\omega_g}{\omega_{ls}} \quad (7)$$

By transferring the generator dynamics to the low-speed side and employing Eqs. (6) and (7), the dynamics of the generator can be expressed as follows

$$n_g^2 J_g \dot{\omega}_g = T_{ls} - (n_g^2 K_g) \omega_g - n_g T_{em} \quad (8)$$

In the case of assuming a perfectly rigid low-speed shaft, it is possible to consider a single mass model for the wind turbine (refer to Fig. 2(b)).

$$J_t \dot{\omega}_r = T_a - K_t \omega_r - T_g \quad (9)$$

where

$$J_t = J_r + n_g^2 J_g$$

$$K_t = K_r + n_g^2 K_g$$

$$T_g = n_g T_{em}$$

## 2.2. Model parameters

The Model parameters used in this research are derived from the CART wind turbine (Controls Advanced Research Turbine) located on the NREL (National Wind Technology Center) site. This originally came from a Westinghouse WTG-600 wind turbine. CART is a two-blade horizontal-axis WT, with a nominal power of 600 kW, with blades 43 m in diameter, and a mast of 35 m [33].

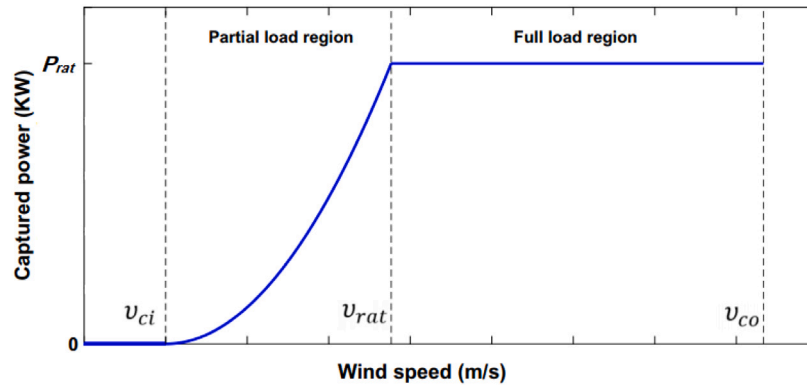


Fig. 3. Wind-speed operating regions of the wind turbine.

### 2.3. Problem formulation

Wind turbine management is essential for the efficient operation of wind energy production systems. In Fig. 3, different control objectives related to the operating zone of the wind turbine are depicted.

The curve  $C_p(\lambda, \beta)$  described in Eq. (1) is unique to each wind turbine and exhibits a distinct maximum value, denoted as  $C_{p_{opt}}$ , at a specific point

$$C_p(\lambda_{opt}, \beta_{opt}) = C_{p_{opt}} \quad (10)$$

In conditions where power production is below the rated power, the  $C_p(\lambda, \beta)$  curve exhibits a unique maximum value. To achieve maximum wind power extraction, the angle of blade pitch  $\beta$  is set to the optimum value  $\beta_{opt}$ , and the wind turbine is designed to operate at its optimal tip-speed ratio  $\lambda_{opt}$ . The main control purpose is the tracking of the optimal rotor speed reference  $\omega_{r_{opt}}$ , which guarantees the maximum capture of wind power.

In the full load region characterized by high wind speeds ( $v_{rat} < v < v_{co}$ ), the main objective is to ensure a steady output power at the rated value and minimize transient loads across all turbine components.

### 2.4. Simulator description (FAST)

The (FAST) code, which stands for Fatigue, Aerodynamics, Structures, and Turbulence respectively, developed by the National Renewable Energy Laboratory (NREL), is an aeroelastic wind turbine (WT) simulator designed for modeling propeller-type machines with two or three blades. This framework is extensively utilized by wind turbine designers to predict fatigue and extreme loads. The simulation approach employed in FAST involves a modal method to model tower components and flexible blades with the remaining elements treated as rigid loads. Within the scope of this study, the simulator incorporates three degrees of freedom (DOFs): the rotor speed (accounting for two DOFs), the variable generator and the blade pitch DOF. The rotor speed and variable generator DOFs accommodate variations in the torsional motion and generator speed between the generator and rotor/hub caused by drive train flexibility. The degree of freedom (DOF) associated with blade teetering considers the motion arising from non-uniform wind forces exerted across the rotor plane [33]. Fig. 4 demonstrates the control scheme through FAST simulator.

## 3. Model-free control of variable speed WT

This section's aim of the paper is to provide a conceptual theoretical framework for the developed controllers. First, Model-Free Controller based Wind Estimator (MFCE) in the below-rated wind speed. Second, a Multi-Variable Model-Free Controller (MV MFC) in the above-rated wind speed.

### 3.1. Model-free control in the low-speed region

This section outlines the utilized approach for achieving maximum power tracking in the below-rated power region, employing a model-free control technique based on wind estimation.

The model-free control approach, developed by Join and Fliess [16], is applied to substitute the complex nonlinear model of the wind turbine presented in Eq. (9) with an ultra-local model:

$$y^{(v)}(t) = F(t) + \beta u(t) \quad (11)$$

For the wind turbine, we set  $y = \omega_r$  and  $u = T_g$  where,

$$\omega_r^{(v)}(t) = F(t) + \beta T_g \quad (12)$$

where,

- $y^{(v)}$  represents the  $v$ th order derivative of the WT observed output.
- The variable  $F$  represents the unmodeled dynamics of the wind turbine plant, encompassing unforeseen disturbances.
- The parameter  $\beta \in \mathbb{R}$  is a non-physical value that is chosen by the designer to ensure compatibility between  $\beta u$  ( $\beta T_g$ ) and  $y^{(v)}$  ( $\omega_r^{(v)}$ ) in terms of their sign and magnitude.

Examining the ultra-local model presented in Eq. (12) again, we can now proceed with the closure of the control loop by incorporating an intelligent PID (iPID) controller:

$$F(t) = \omega_r^{(v)}(t) - \beta T_g \quad (13)$$

The control signal is expressed consequently:

$$T_g = -\frac{F - \omega_{r_{opt}}^{(v)} + u_c}{\beta} \quad (14)$$

where,

- The variable  $\omega_{r_{opt}}^{(v)}$  corresponds to the  $v$ th derivative of the WT output reference trajectory.
- The variable  $\omega_r^{(v)}$  corresponds to the  $v$ th derivative of the WT output trajectory.

The combination of Eqs. (13) and (14) results in the following expression:

$$\omega_r^{(v)} = F + \beta \left( -\frac{F - \omega_{r_{opt}}^{(v)} + u_c}{\beta} \right) = \omega_{r_{opt}}^{(v)} - u_c \quad (15)$$

Afterwards,

$$\omega_r^{(v)} - \omega_{r_{opt}}^{(v)} + u_c = 0 \quad (16)$$

$$e^{(v)} + u_c = 0 \quad (17)$$



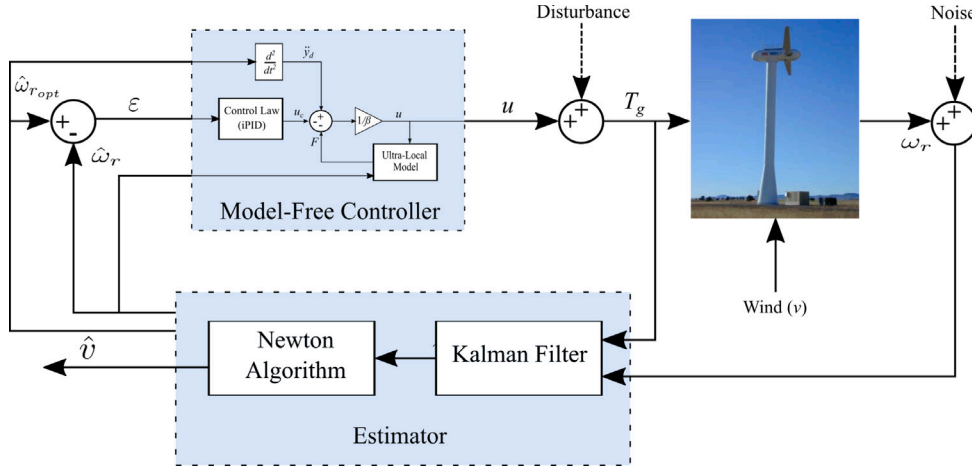


Fig. 4. Model-free controller based wind estimator scheme (FAST).

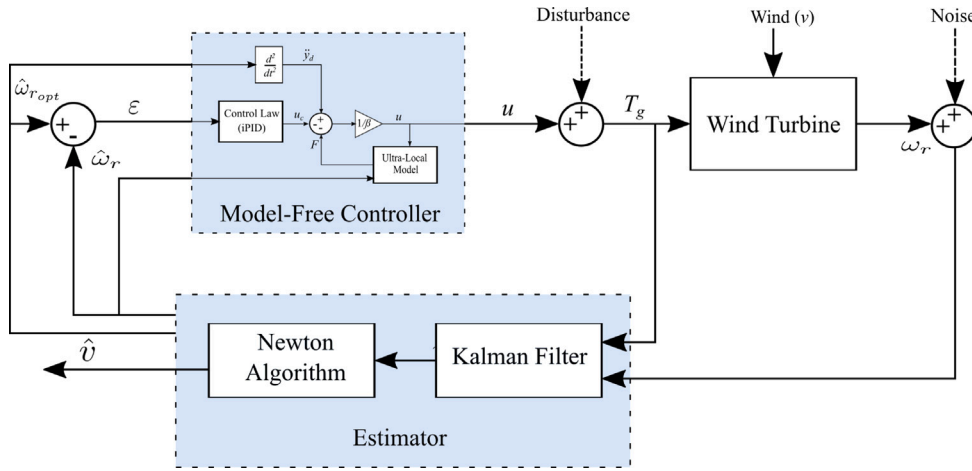


Fig. 5. Model-free controller based wind estimator scheme.

In which,  $e^{(v)}$  represents the  $v$ th derivative of the error ( $e = \omega_r - \omega_{r_{opt}}$ ). The selection of the control signal  $u_c$  is imperative to ensure exact asymptotic tracking, whereby the output effectively converges to the reference trajectory, that means,

$$\lim_{t \rightarrow +\infty} e(t) = 0 \quad (18)$$

For the Intelligent PID setup, the control signal in Eq. (14) is determined through a conventional PID controller,

$$u = T_g = -\frac{F - \ddot{\omega}_{r_{opt}} + K_P + K_I \int e + K_D \dot{e}}{\beta} \quad (19)$$

Here, we consider  $v = 2$  in Eq. (12) based on both practical experience and the physical characteristics of wind turbine dynamics. By substituting Eq. (14) into Eq. (19), the intelligent proportional-integral-derivative controller (iPID) can be described as follows:

$$\ddot{e} + K_P e + K_I \int e + K_D \dot{e} = 0 \quad (20)$$

The design process is carried out in two steps, as depicted in Fig. 5. Initially, the wind turbine's ultra-local model is estimated. Subsequently, the aerodynamic torque is estimated, and finally, the wind speed is derived.

### 3.1.1. Estimation of the wind turbine's ultra-local model

Eq. (12) is analyzed by employing the fundamental principles of operational calculus to encompass all the preceding equations. Within

this framework, the variable  $F$  is substituted with  $F_{approx}$ , which is considered to retain a constant value throughout the interval  $[t, t + T]$ . The determination of  $F_{approx}$  involves a sequence of steps as delineated below:

1. Applying the Laplace transformation to Eq. (12) with  $v = 2$  yields:

$$s^2 \Omega_r(s) - s\omega_r(0) - \dot{\omega}_r(0) = \frac{F_{approx}}{s} + \beta U(s) \quad (21)$$

2. Taking the derivative of Eq. (21) with respect to  $s$  and evaluating at  $s = 0$ , we can eliminate  $\dot{\omega}_r(0)$ . This yields:

$$2\Omega_r(s) + s^2 \frac{d}{ds} \Omega_r(s) - \omega_r(0) = -1s^{-2} F_{approx} + \beta \frac{d}{ds} U(s) \quad (22)$$

3. By differentiating once more with respect to  $s$  and evaluating at  $s = 0$ , we can eliminate  $\omega_r(0)$ . This results in:

$$2\Omega_r(s) + 4s \frac{d}{ds} \Omega_r(s) + s^2 \frac{d^2}{ds^2} \Omega_r(s) = 2s^{-3} F_{approx} + \beta \frac{d^2}{ds^2} U(s) \quad (23)$$

4. To eliminate the time derivatives and apply filtering, we multiply Eq. (23) by  $\frac{1}{s^3}$  and ensure that each factor is integrated at least once. This yields the following expression:

$$2 \frac{1}{s^3} \Omega_r(s) + 4 \frac{1}{s^2} \frac{d}{ds} \Omega_r(s) + \frac{1}{s} \frac{d^2}{ds^2} \Omega_r(s) = 2F_{approx} \frac{1}{s^6} + \beta \frac{1}{s^3} \frac{d^2}{ds^2} U(s) \quad (24)$$

5. By recalling that differentiation of  $\Omega_r(s)$  according to  $s$  corresponds to multiplication by  $-t$  in the time domain, and that left multiplying  $\Omega_r(s)$  by  $\frac{1}{s^\alpha}$ , where  $\alpha \geq 1$ , represents iterated integrals, we can use the Cauchy formula for repeated integration to obtain a closed-form expression for the operator  $\frac{1}{s^\alpha} \frac{d^n}{ds^n}$  in terms of a single integral:

$$\frac{1}{s^\alpha} \frac{d^n}{ds^n} \Omega_r(s) \longleftrightarrow \frac{(-1)^n}{(\alpha-1)!} \int_0^t (t-\tau)^{\alpha-1} \tau^n \omega_r(\tau) d\tau \quad (25)$$

6. Eq. (25) yields the following time-domain representation expression:

$$F_{\text{approx}} = \frac{60}{t^5} \int_0^t (t^2 - 6t\tau + 6\tau^2) \omega_r(\tau) d\tau - \frac{30\beta}{t^5} \int_0^t (t-\tau)^2 \tau^2 u(\tau) d\tau \quad (26)$$

The expression presented above can be effectively implemented in a digital system as a Finite Impulse Response (FIR) filter, as documented in the work by Sehili et al. [30]. To achieve this objective, the method is adapted by incorporating a backward integration scheme within a narrow time window of fixed duration  $T$ , enabling a viable real-time implementation. Following appropriate variable transformations, the resulting equation, represented by Eq. (27), represents the ultimate form of the algebraic estimator for approximating  $F_{\text{approx}}$ :

$$F_{\text{approx}} = \frac{60}{T^5} \int_0^T (T^2 - 6T\tau + 6\tau^2) \omega_r(t-\tau) d\tau - \frac{30\beta}{T^5} \int_0^T (T-\tau)^2 \tau^2 u(t-\tau) d\tau \quad (27)$$

This modified approach ensures practical feasibility and facilitates the deployment of the estimator in real-time control systems.

### 3.1.2. Aerodynamic torque estimation

The wind speed, denoted as  $v$ , in the equations related to the aerodynamic torque and power, represents an average value of the three-dimensional wind speed field that affects the rotor blades. This wind speed field exhibits spatial variations across the swept area of the rotor, making it challenging to characterize with a single measure. As a result, a unique representation of the wind speed field is not possible [34].

A Kalman filter is employed for the estimation of the aerodynamic torque  $T_a$ ,

$$\begin{bmatrix} \dot{\omega}_r \\ \dot{T}_a \end{bmatrix} = \begin{bmatrix} -\frac{K_t}{J_r} & \frac{1}{J_r} \\ 0 & 0 \end{bmatrix} \begin{bmatrix} \omega_r \\ T_a \end{bmatrix} + \begin{bmatrix} -\frac{1}{J_r} \\ 0 \end{bmatrix} T_g + \begin{bmatrix} 0 \\ 1 \end{bmatrix} \xi$$

$$y = \omega_r + \zeta \quad (28)$$

The process noise  $\xi$  and measurement noise  $\zeta$  are incorporated into the system as centered white, non-correlated Gaussian noise. These noise components account for random fluctuations and uncertainties in the system. Despite the non-zero variance of the process noise, the Kalman estimator for the system described by Eq. (28) exhibits convergence. In the Kalman filter technique, the approximated state  $[\hat{\omega}_r, \hat{T}_a]^T$  is not directly used to determine the ideal reference speed  $\omega_{r_{opt}}$ .

By utilizing Eq. (14) alongside the estimated values of the rotor speed and aerodynamic torque obtained from Eq. (28), the wind speed estimation  $\hat{v}$  can be determined through the subsequent algebraic equation,

$$\hat{T}_a - \frac{1}{2} \rho \pi R^3 C_q \left( \frac{\hat{\omega}_r R}{\hat{v}} \right) \hat{v}^2 = 0 \quad (29)$$

By employing the Newton algorithm, Eq. (29) is effectively solved. Upon obtaining the estimated wind speed ( $\hat{v}$ ), the corresponding estimated optimal rotor speed  $\hat{\omega}_{r_{opt}}$  can be readily determined,

$$\hat{\omega}_{r_{opt}} = \frac{\lambda_{opt} \hat{v}}{R} \quad (30)$$

Subsequently, the estimator is employed in the development of nonlinear control laws that facilitate optimal tracking of the rotor speed  $\omega_{r_{opt}}$ .

### 3.1.3. Performance criteria

The design of an optimal controller is guided by a set of performance criteria, ensuring that the controller meets the desired objectives.

1. The electrical efficiency denoted by  $\eta_{\text{elec}}$ , is the ratio of electrical energy produced during the simulation to the maximum available aerodynamic energy. It quantifies the wind turbine's effectiveness in converting aerodynamic energy into electrical energy.

$$\eta_{\text{elec}} = \frac{\int_{t_{\text{ini}}}^{t_{\text{end}}} P_e dt}{\int_{t_{\text{ini}}}^{t_{\text{end}}} P_{a_{opt}} dt} \quad (31)$$

such that, the aerodynamic efficiency  $\eta_{\text{aero}}$  represents the ratio of aerodynamic energy extracted by the WT through the simulation of the maximum available aerodynamic energy.

$$\eta_{\text{aero}} = \frac{\int_{t_{\text{ini}}}^{t_{\text{end}}} P_a dt}{\int_{t_{\text{ini}}}^{t_{\text{end}}} P_{a_{opt}} dt} \quad (32)$$

where,

$$P_{a_{opt}} = \frac{1}{2} \rho \pi R^2 C_{p_{opt}} v^3 \quad (33)$$

2. The second criterion involves examining the statistical characteristics of the control signal, specifically the mean, maximum, and standard deviation values.

## 3.2. Model-free control in the high-speed region

For high wind speed conditions, the main goal is to sustain a consistent output power at its rated magnitude. This is typically accomplished by keeping a constant generator torque while adjusting the blade pitch angle to achieve a steady rotor speed. To address this control objective, a novel multi-variable model-free control approach is proposed, which leverages both generator torque and blade pitch angle as control inputs. This innovative control strategy aims to optimize the performance and stability of the wind turbine in the presence of elevated-speed wind conditions.

### 3.2.1. Estimation of the wind turbine's ultra-local models

To address the power-tracking error of the torque controller, a sub-system with single-input, single-output (SISO) model-free (MF) controller dynamics is introduced. This sub-system is designed to regulate and minimize the power-tracking error, ensuring accurate and efficient power control such that,

$$e_1 = P_{\text{nom}} - P_e \quad (34)$$

To support the torque controller acting in the electric power regulation output of the wind turbine, while mitigating excessive loads and ensuring the rotor speed remains within permissible limits, an additional input is incorporated into the model-free controller. This supplementary input is derived from the rotor speed tracking error, thereby contributing to the effective operation of the control system,

$$e_2 = \omega_{\text{nom}} - \omega_t. \quad (35)$$

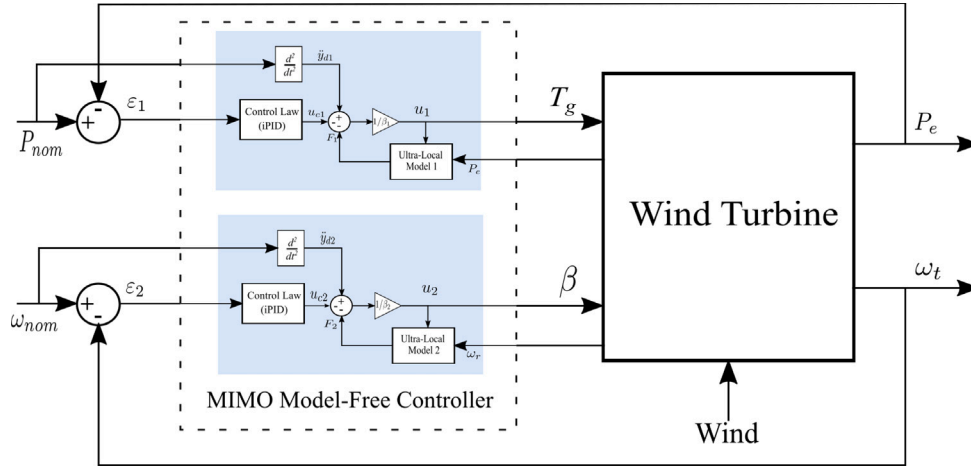


Fig. 6. MV Model-free control design scheme.

The extended multi-variable ultra local-model is presented by the subsequent equation:

$$y_i^{(n_i)}(t) = F_i(t) + \beta_i u_i(t) \quad (36)$$

In the given equation,  $i$  represents the number of variables, while  $n_i$  denotes the order derivative of the observed output. As discussed in Section 3.1, in the case of multiple variables, both  $n_1$  and  $n_2$  are set to 2.

$$\begin{aligned} y_1^{(n_1)}(t) &= F_1(t) + \beta_1 u_1(t) \\ y_2^{(n_2)}(t) &= F_2(t) + \beta_2 u_2(t) \end{aligned} \quad (37)$$

By considering Eqs. (34) and (35) and referring to the wind turbine system depicted in Fig. 6, the multi-variable ultra-local model stated in Eq. (37) can be represented as follows:

$$\begin{aligned} \ddot{P}_e(t) &= F_1(t) + \beta_1 T_g(t) \\ \ddot{\omega}_t(t) &= F_2(t) + \beta_2 \beta(t) \end{aligned} \quad (38)$$

The control signals can be expressed using Eqs. (39) and (40):

$$u_1 = T_g = -\frac{F_1 - \ddot{P}_{e_{nom}} + u_{c1}}{\beta_1} \quad (39)$$

$$u_2 = \beta = -\frac{F_2 - \ddot{\omega}_{t_{nom}} + u_{c2}}{\beta_2} \quad (40)$$

where,

- $\ddot{P}_{e_{nom}}$  and  $\ddot{\omega}_{t_{nom}}$  represent the second derivative of the output reference trajectory.
- $u_{c1}$  and  $u_{c2}$  denote the causal feedback control signals.
- $\beta_1$  and  $\beta_2$  represent non-physical parameters that are chosen by the designer.
- $F_1$  and  $F_2$  refer to the un-modeled dynamics of the wind turbine plant.
- $T_g$  represents the generator torque, while  $\alpha^1$  denotes the pitch angle.

The estimation of  $F_1$  and  $F_2$  is obtained using the same procedure outlined in the previous section, ensuring that the accuracy and

reliability of the estimation process are maintained such that,

$$\begin{aligned} F_{1_{approx}} &= \frac{60}{t^5} \int_0^t (t^2 - 6t\tau + 6\tau^2) P_e(\tau) d\tau - \frac{30\beta_1}{t^5} \int_0^t (t - \tau)^2 \tau^2 T_g(\tau) d\tau \\ F_{2_{approx}} &= \frac{60}{t^5} \int_0^t (t^2 - 6t\tau + 6\tau^2) \omega_t(\tau) d\tau - \frac{30\beta_2}{t^5} \int_0^t (t - \tau)^2 \tau^2 \alpha(\tau) d\tau \end{aligned} \quad (41)$$

Although the wind speed varies continuously, the estimated terms  $F_1$  and  $F_2$  appear nearly constant because the control objective in the above-rated region is to regulate rotor speed and electrical power around fixed nominal values. Given the high tracking precision of the proposed controller, both the system outputs and control inputs remain stable, resulting in quasi-constant estimates of the ultra-local model terms.

### 3.2.2. Performance criteria

The optimal controller is chosen by considering the following performance criteria:

- Maintaining the electrical power  $P_e$  in close proximity to the nominal power  $P_{nom}$ .
- Regulating the rotor speed  $\omega_t$  to track the nominal trajectory with minimal deviations.
- Ensuring that the blade pitch angle remains within acceptable operating limits.

## 4. Simulation results

The wind turbine utilized for the numerical simulations is described by the parameters presented in Table 1. These specific parameters are associated with the Controls Advanced Research Turbine (CART), situated at the National Wind Technology Center of the National Renewable Energy Laboratory (NREL) near Boulder, CO. The CART is identified as a wind turbine featuring variable-speed and variable-pitch capabilities, with a designated hub height of 36 m and a nominal power capacity of 600 kW. It is characterized by a two-bladed, teetered hub configuration with a diameter of 43 m [33].

To facilitate a thorough comparison, the turbine was simulated using two different approaches: a mathematical model and the aeroelastic simulator FAST.

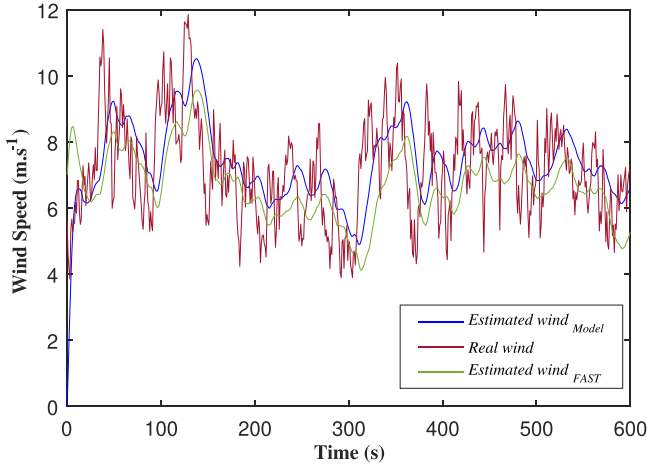
### 4.1. Results of MFC below the rated-power

The wind input employed in the simulations encompasses a comprehensive turbulent wind field that encompasses the entire computational

<sup>1</sup> To maintain consistency with standard wind energy literature, the blade pitch angle is denoted by  $\beta$ . The symbols  $\beta_1$  and  $\beta_2$ , on the other hand, refer to the gain parameters used in the ultra-local models of the form  $\dot{y}_i = F_i + \beta_i u_i$ . The distinction between these uses is made clear by their subscripts and context throughout the manuscript.

**Table 1**  
Wind turbine and controller design parameters.

| Parameter                               | Value    |
|-----------------------------------------|----------|
| Gearbox ratio                           | 43.165   |
| Rotor diameter                          | 43.3 m   |
| Hub height                              | 36.6 m   |
| Maximum rotor torque                    | 162 kN m |
| Electrical power generator system       | 600 kW   |
| Sampling time ( $T_s$ )                 | 0.0001 s |
| Estimation window ( $T$ )               | 0.2 s    |
| Samples per window ( $N$ )              | 2000     |
| Controller type                         | iPD      |
| MFCE tuning (Below Rated)               |          |
| Pole location ( $\lambda$ )             | -10      |
| Proportional gain ( $K_P = \lambda^2$ ) | 100      |
| Derivative gain ( $K_D = -2\lambda$ )   | 20       |
| MV-MFC tuning (Above Rated)             |          |
| $\lambda_1$ (Power)                     | -80      |
| $\beta_1$ (Power loop gain)             | 1000     |
| $K_{P1} = \lambda_1^2$                  | 6400     |
| $K_{D1} = -2\lambda_1$                  | 160      |
| $\lambda_2$ (Pitch)                     | -40      |
| $\beta_2$ (Pitch loop gain)             | 40       |
| $K_{P2} = \lambda_2^2$                  | 1600     |
| $K_{D2} = -2\lambda_2$                  | 80       |



**Fig. 7.** Wind speed profile of 7 m s<sup>-1</sup> mean value — Kaimal spectrum.

domain. Fig. 7 visualizes the temporal variation of wind speed recorded at the hub height. The instantaneous wind speed at a specific point, represented as  $v$ , is formed by the combination of two distinct components:

$$v = v_t + v_m \quad (42)$$

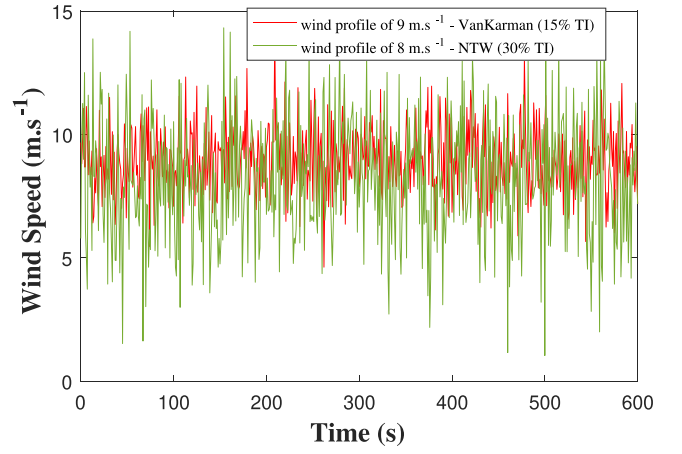
Such that,  $v_m$  represents the mean value, while  $v_t$  represents the turbulent component of the wind speed.

The experimental wind spectra by Van der Hoooven reveal that the mean value  $v_m$  peaks at an average period of 10 min. To ensure a consistently steady mean value, a wind dataset spanning 10 min is specifically chosen for analysis [32].

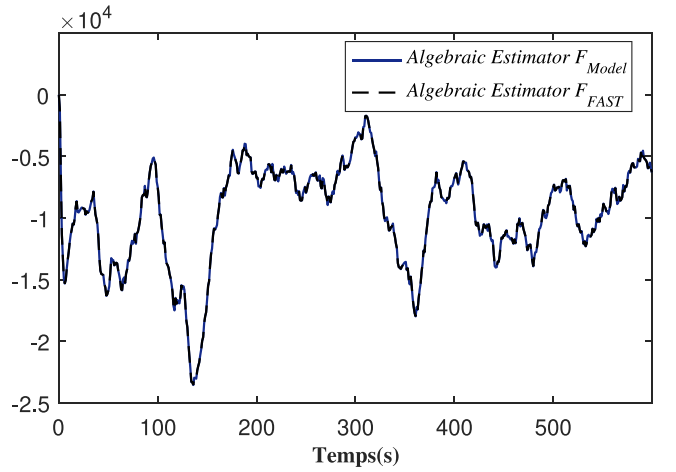
The turbulent wind data utilized in the simulations were generated using class A Kaimal turbulence spectra. At the hub height, the data feature a turbulence intensity of 25% and a mean value of 7 m/s.

#### 4.1.1. Results validation using the simplified mathematical model and FAST simulator

The study was conducted using a sampling frequency of 10 kHz. To estimate the ultra-local model parameter  $F$  as outlined in Eq. (13), a fixed sampling interval of  $T_s = 10^{-4}$  s, an estimation window of  $T = 0.2$  s, and a sample count  $N = 2000$  were applied.



**Fig. 8.** Wind speed profiles — Van Karman and NTW spectra.



**Fig. 9.** Algebraic estimation of the unmodeled dynamics of the CART WT.

In the forthcoming simulations, an i-PD controller is employed. Through the selection of  $K_P = \lambda^2$  and  $K_D = 2\lambda$ , where  $\lambda \in \mathbb{R}^+$ , the closed loop is stable with 2 real negative poles corresponding to  $-\lambda$ :

$$\ddot{e} + \lambda^2 e + 2\lambda \dot{e} = 0 \quad (43)$$

The wind turbine (WT) system experiences an additional disturbance  $d$  in the control torque with a magnitude of 10 kN m and measurement noise signal-to-noise ratio (SNR) in  $\omega_g$  of approximately 7 dB.

As previously mentioned, the control design objective is to capture the maximum amount of wind energy and mitigate transient loads on the WT.

Fig. 10 depicts the numerical results obtained from the proposed Model-free controller employing the simplified mathematical model, along with the validation results obtained using the FAST simulator. The primary goal of the controller is to guarantee that the rotor speed  $\omega_r$  closely follows the average behavior of the optimal rotor speed  $\omega_{r,opt}$ , while disregarding the turbulent part. The estimation of the unmodeled dynamics  $F$  begins the online update at the initial time, as shown in Fig. 9. The control effort signal, presented in Fig. 11, demonstrates that the controlled torque remains below the maximum design limit of 162 kNm. Additionally, the generator torque exhibits smooth behavior, resulting in low-frequency oscillations in the generator currents. Favorable conditions such as these contribute to the optimal performance of electrical devices.



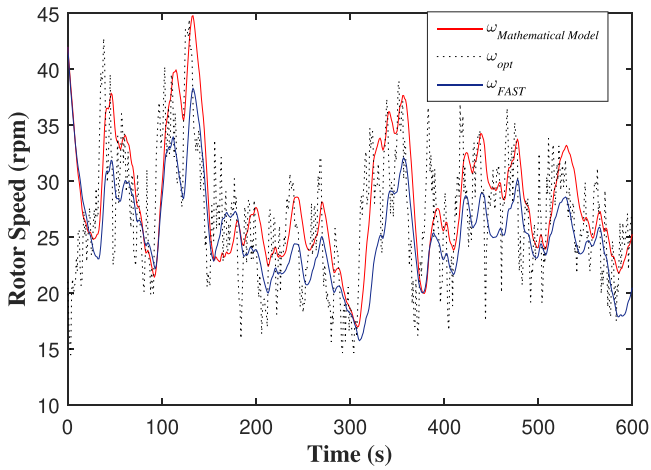


Fig. 10. Model-free controller based wind estimator: Rotor speed responses.

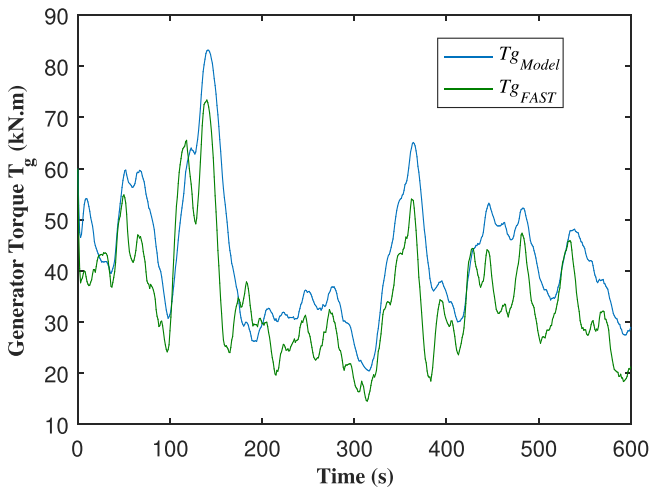


Fig. 11. Model-free controller based wind estimator: Generator torque.

The combination Kalman filter and Newton algorithm yields an accurate estimation of the wind speed (illustrated in Fig. 7). This estimation is achieved by utilizing the estimates of the rotor speed and the aerodynamic torque, even in the presence of measurement noises in  $T_g$  and  $\omega_r$ .

Furthermore, it can be observed that the electrical power generated by the wind turbine simulator FAST is lower than that generated by the simplified mathematical model (as shown in Fig. 12). The reduction in power output can be attributed to the greater complexity of the WT simulator, which incorporates additional flexible elements and subsequently incurs additional energy losses compared to the simplified mathematical model. The performance of the controller is evaluated and presented in Tables 2 and 3.

The robustness of the proposed model-free control strategy was assessed under different turbulent wind profiles. Simulations were conducted using wind inputs generated from Kaimal, von Kármán, and NWT spectra, each representing different turbulence characteristics. The Kaimal spectrum featured a mean wind speed of 7 m/s with 25% turbulence intensity, while the von Kármán and NWT spectra had mean wind speeds of 9 m/s and 8 m/s, with turbulence intensities of 15% and 30%, respectively.

The used wind profiles are illustrated in Fig. 8, and the corresponding rotor speed responses are presented in Fig. 13. Updated performance metrics under these different turbulence conditions are summarized in Table 2. Results confirm that the controller consistently

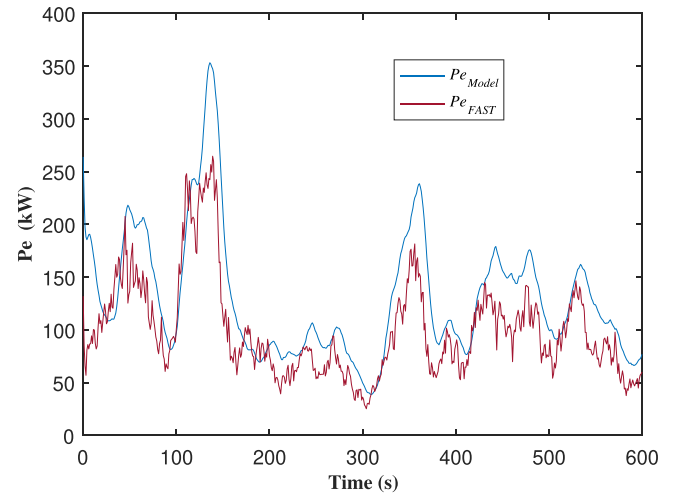


Fig. 12. Model-free controller based wind estimator: WT electrical power.

Table 2

MFC performances using mathematical model and comparison with other controllers.

|                             | Efficiency (%) | Standard deviation of $T_g$ (kN m) | max ( $T_g$ ) (kN m) |
|-----------------------------|----------------|------------------------------------|----------------------|
| ISC                         | 80.32          | 13.63                              | 124.02               |
| ATF                         | 73.04          | 14.56                              | 77.90                |
| NSSFE                       | 84.54          | 14.12                              | 82.00                |
| MFCE <sub>(kaimal)</sub>    | 91.30          | 12.87                              | 83.14                |
| MFCE <sub>(vonkarmán)</sub> | 92.45          | 3.85                               | 72.67                |
| MFCE <sub>(nwt)</sub>       | 84.66          | 6.64                               | 73.56                |

Table 3

MFC performances using FAST and comparison with other controllers.

|       | Efficiency (%) | Standard deviation of $T_g$ (kN m) | max ( $T_g$ ) (kN m) |
|-------|----------------|------------------------------------|----------------------|
| ISC   | 51.79          | 13.14                              | 112.34               |
| ATF   | 63.43          | 15.02                              | 76.99                |
| NSSFE | 61.23          | 11.91                              | 70.70                |
| MFCE  | 73.44          | 11.49                              | 73.02                |

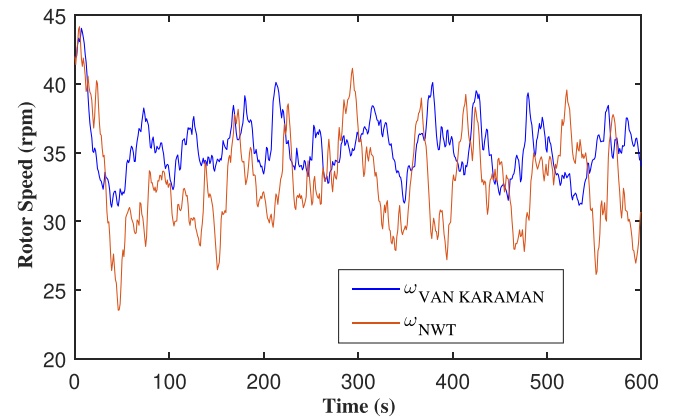


Fig. 13. Model-free controller based wind estimator: Rotor speed responses under different wind conditions.

maintained stability and effective tracking performance across varying turbulence intensities and spectral distributions, demonstrating strong robustness against different wind environments.

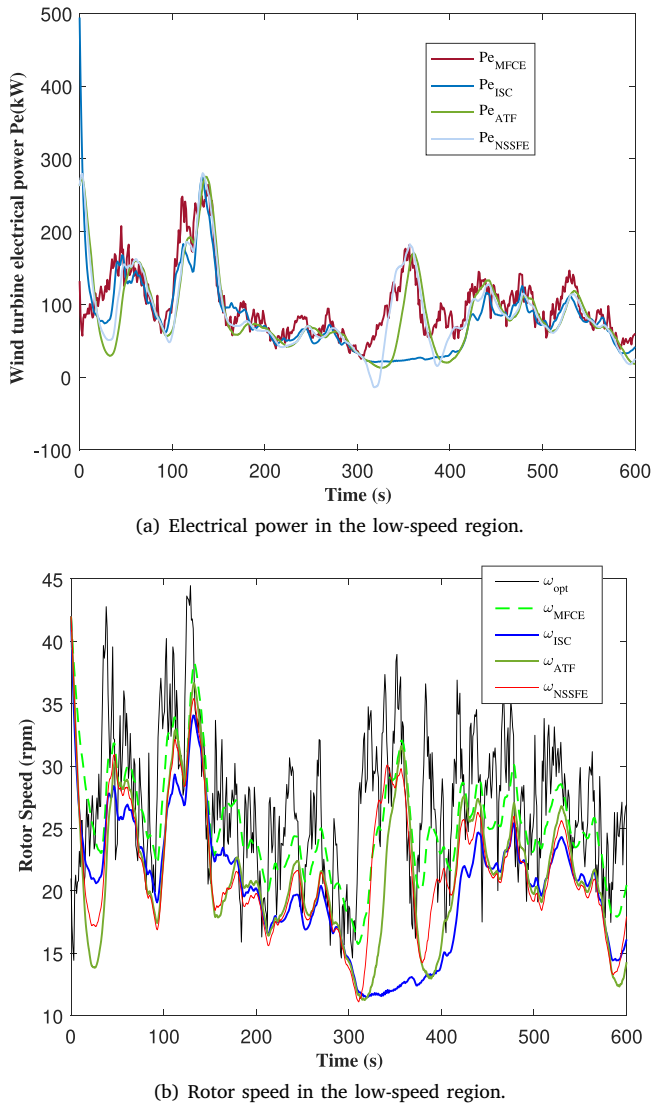


Fig. 14. Comparison between different controllers in the low-speed region.

#### 4.1.2. Comparison with existing control strategies

The proposed model-free control based wind estimator approach has proven to be inadequate in achieving the desired control objectives of the wind turbine under low wind conditions in real operating conditions, particularly in the presence of disturbances and noise measurements.

The experimental evidence of the proposed controller's performance is now examined by comparing it with existing classical control techniques. Tables 2 and 3, along with Figs. 14(a) and 14(b), present a comparison of the performance achieved using the proposed model-free control based wind estimator (MFCE) with different control approaches under the same simulation conditions. The compared techniques include Indirect speed control (ISC), Aerodynamic torque feed forward control (ATF), and Nonlinear static state feedback control with estimator (NSSFE) [7].

The proposed controller demonstrates significantly improved tracking of the optimal rotor speed  $\omega_{r_{opt}}$  compared to classically designed controllers. This is evident in Figs. 10 and 14(b), where the presence of the disturbance  $d$  does not affect the accurate tracking of  $\omega_{r_{opt}}$ . The rotor speed  $\omega_r$  effectively follows the average trend of the optimal rotor speed, without being affected by rapid variations in the reference signal, that is obtained from the turbulent wind speed.

All of these achievements, as previously mentioned, are made possible through the estimation of the parameter  $F$ , which encompasses all the dynamics of the system. This estimation enables a rapid convergence of the error, striking a good balance between wind energy capture and load reduction.

The rotor consistently captures aerodynamic power that closely aligns with the optimal aerodynamic efficiency curve,  $P_{a_{opt}}$ . The electrical efficiency, which exceeds 91% using the mathematical model and 73% using FAST, is relatively unaffected by measurement noise and disturbances. The control mechanism effectively mitigates the impact of disturbances, contributing to the maintenance of high electrical efficiency. The smooth variation in rotor and generator speed also results in reduced friction losses.

The performances of the proposed controller are found to be almost identical to the case where wind speed and other system variables were assumed to be accessible. The control forces demonstrate acceptability, exhibiting a significantly low standard deviation when compared to other controllers.

As depicted in Fig. 11 and summarized in Tables 2 and 3, the transient forces on the drive device remain acceptable. The precise tracking of the optimal aerodynamic efficiency curve is evident, resulting in notably higher electrical power production by the wind turbine compared to all previously used controllers.

The application of the mathematical model resulted in noteworthy enhancements in electrical power efficiency, with improvements of 11%, 18%, and 8% achieved for the ISC, ATF, and NSSFE techniques, respectively. Intriguingly, when evaluated using the FAST simulator, the proposed MFCE exhibits compelling evidence of even greater progress, achieving increases in electrical power efficiency by 22%, 12%, and 10% over the ISC, ATF, and NSSFE techniques, respectively.

Furthermore, comparison with observer-based control methods which have been widely studied and applied in wind turbine systems to estimate unmeasured states or compensate for disturbances. These include fuzzy observers with adaptive backstepping control [35], high-order sliding mode observers [36], and observer-based robust fault predictive control [37]. While effective under well-modeled conditions, these methods depend heavily on the availability of accurate system dynamics and sensor information and the performance was sensitive to model uncertainty and required gain scheduling.

In contrast, the proposed model-free control (MFC) strategy eliminates the need for model-based observers entirely. Using ultra-local models and algebraic estimation, the MFC captures the system's dynamic behavior in real time directly from input-output data, without requiring aerodynamic coefficients or system linearization.

From the previous discussion, it is evident that the MFC-based wind estimator demonstrates a strong performance in terms of efficiency, reduction of control and transients experienced by the wind turbine, as besides effective disturbance rejection even in the presence of noise measurements.

#### 4.2. Results of MFC above the rated power

The simulations employ a 10-minute dataset of full-field turbulent wind as the wind inflow. Fig. 15 depicts the variation of the wind speed at hub-height. The turbulent wind data utilized in the simulations were generated using class A Kaimal turbulence spectra. At the hub height, the data feature a turbulence intensity of 15% and a mean value of 20 m/s. This wind scenario represents the operational constraints of the turbine. In the context of this excitation, the discussed controllers are individually assessed to optimize the wind turbine's operational duration while effectively managing output power regulation, rotor speed, and reducing transient loads.

##### 4.2.1. Results validation using the simplified mathematical model and FAST simulator

The examination of the model-free multivariable controller on both the mathematical model and FAST has revealed the possibility of

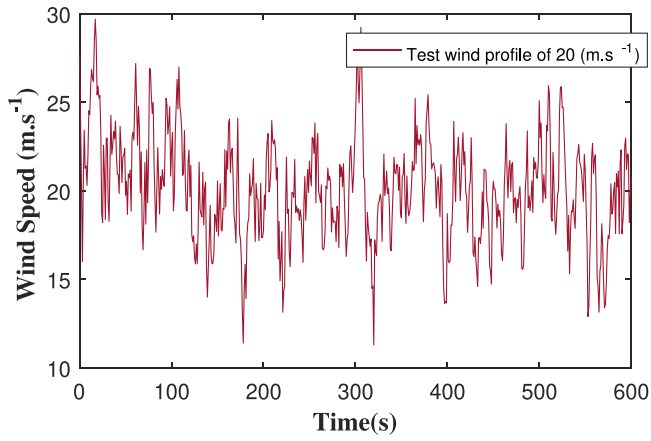
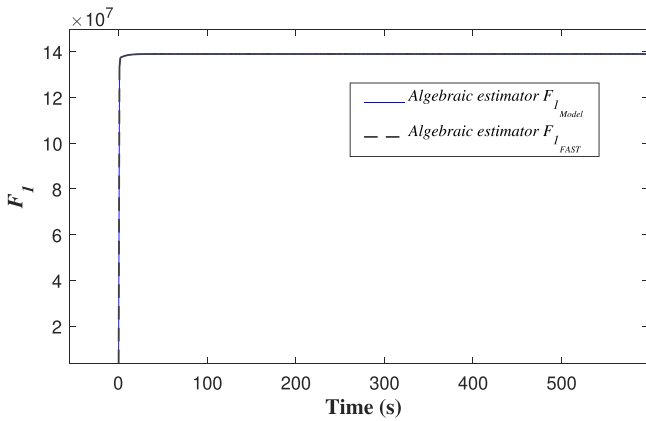
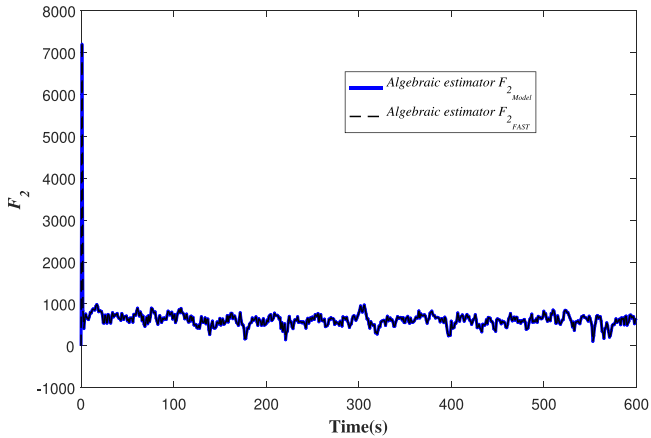


Fig. 15. Wind speed profile of 20 ( $\text{m s}^{-1}$ ) mean value.



(a) Estimation of the first sub-system.



(b) Estimation of the second sub-system.

Fig. 16. Algebraic estimation of the unmodeled dynamics of the CART WT.

achieving a highly satisfactory balance between the regulation of both outputs, all the while maintaining acceptable control loads.

The iPID parameters for the first sub-system of power regulation are set as follows:  $\beta_2$  and  $\lambda_2$  (pole location) are tuned to be equal to 1000 and 80, respectively. For the second sub-system of blade pitch angle control,  $\beta_2$  and  $\lambda_2$  are tuned to be equal to 40 and 40, respectively. The algebraic estimators are presented in Figs. 16(a) and 16(b).

The proposed multi-variable Model-free control approach demonstrates an excellent balance between power and rotor speed regulation,

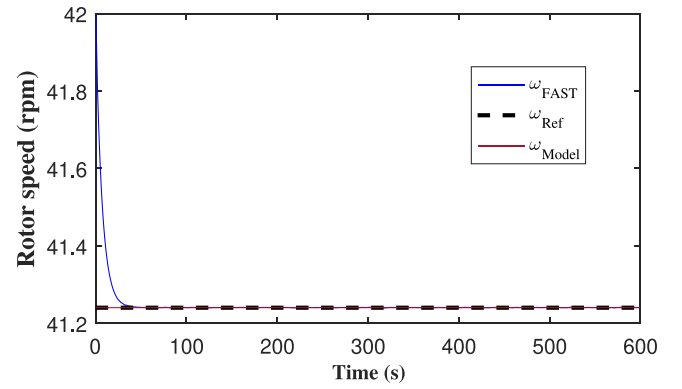


Fig. 17. MV Model-free controller: Rotor speed responses.

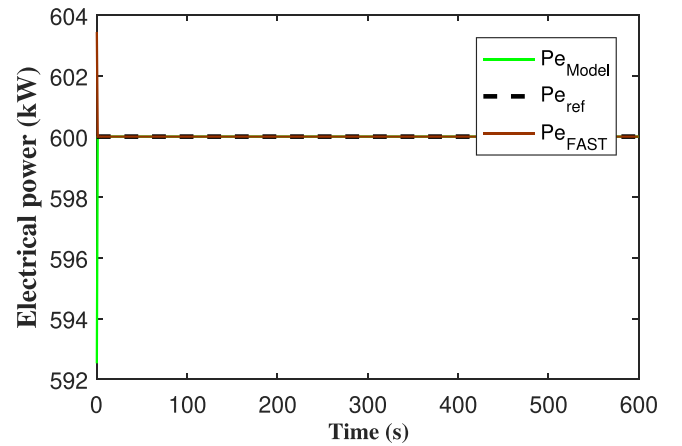


Fig. 18. MV Model-free controller: Electrical power regulation.

Table 4

MV MFC performances using mathematical model and comparison with other controllers.

|                          | PID    | LQG    | MV(NL PID) | MV MFC     |
|--------------------------|--------|--------|------------|------------|
| $\bar{P}_e$ (kW)         | 591.97 | 592.38 | 600.06     | 600.00     |
| std ( $P_e$ ) (kW)       | 32.02  | 26.51  | 0.6222     | $5.25e-07$ |
| std ( $\omega_r$ ) (rpm) | 2.23   | 1.84   | 1.40       | $3.70e-04$ |
| std ( $T_g$ ) (kN m)     | –      | –      | 4.88       | 0.0012     |
| std ( $\beta$ ) (deg)    | 2.39   | 3.78   | 3.38       | $3.70e-04$ |

Table 5

MV MFC performances using FAST and comparison with other controllers.

|                          | PID    | LQG    | MV(NL PID) | MV MFC                |
|--------------------------|--------|--------|------------|-----------------------|
| $\bar{P}_e$ (kW)         | 592.03 | 592.46 | 600.01     | 600.00                |
| std ( $P_e$ ) (kW)       | 45.80  | 31.17  | 0.0454     | $3.15 \times 10^{-5}$ |
| std ( $\omega_r$ ) (rpm) | 3.19   | 2.17   | 1.51       | $1.41 \times 10^{-4}$ |
| std ( $T_g$ ) (kN m)     | –      | –      | 4.35       | $4.34 \times 10^{-3}$ |
| std ( $\beta$ ) (deg)    | 3.40   | 4.63   | 3.84       | $2.68 \times 10^{-4}$ |

which is evident from Figs. 17 and 18. The rotor speed is effectively maintained around its nominal value, with minimal standard deviation in the control signals. The performance results, detailing rotor speed regulation and control signal standard deviation, are summarized in Tables 4 and 5.

The power regulation performance is deemed satisfactory, as demonstrated in Fig. 18, where  $P_e$  consistently persists in a very close proximity to its nominal value, both in the mathematical model and FAST simulator. The control signals, which encompass pitch action and control torque, are effectively illustrated in Fig. 19.

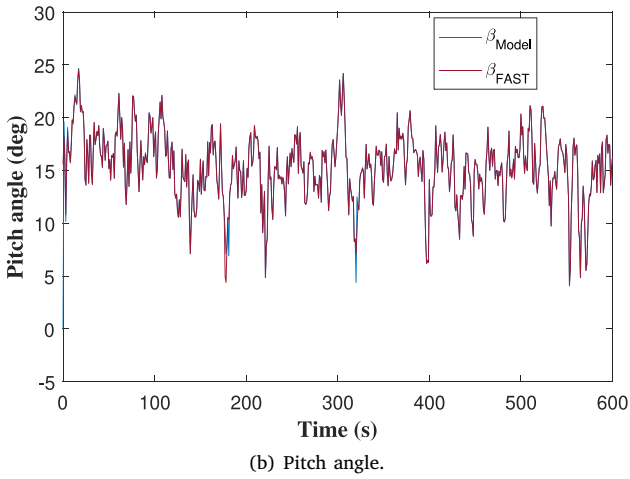
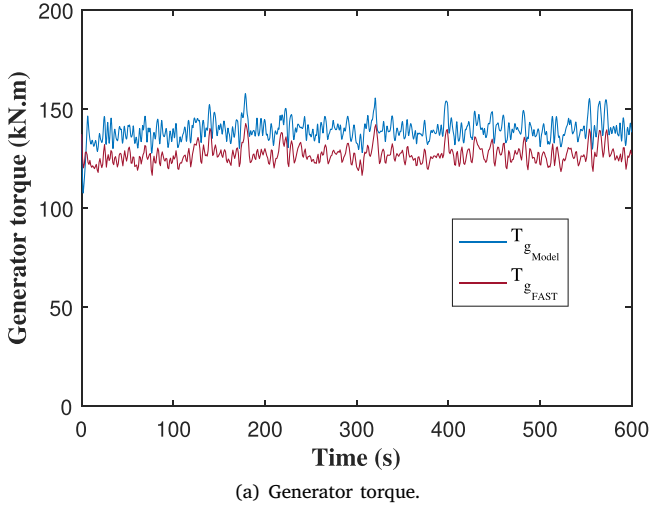


Fig. 19. MV Model-free controller: Control signals.

#### 4.2.2. Comparison with existing control strategies

To emphasize the advantages of employing a multivariable model-free control approach for torque and pitch, we conducted a comprehensive comparison with various classical control strategies applied to both the mathematical model and FAST simulator, particularly focusing on power regulation above the nominal level. The compared approaches encompass Non-linear design, PID, and LQG, as documented in the work by Boukhezzar et al. [38]. By contrasting the performance of these conventional control methods with the proposed model-free multivariable control, we can gain valuable insights into the superiority and effectiveness of the latter in achieving enhanced power regulation and control stability.

It is evident from Fig. 20(b) that the rotor and generator speeds are excellently regulated around their nominal values, specifically at 41 rpm, using the MIMO MF controller. Additionally, Tables 4 and 5 reveal that the standard deviation of  $\omega_r$  remains remarkably low, at  $1.41 \times 10^{-4}$  rpm, underscoring the precision of the speed regulation. Moreover, the electrical power regulation is effectively ensured, as depicted in Fig. 20(a), with a minute standard deviation of  $3.15 \times 10^{-4}$  rpm.

In terms of control efforts, the torque action  $T_g$  consistently remains below the acceptable value of 162 kN m, and its standard deviation is also impressively low, measuring  $6.31 \times 10^{-9}$ . Furthermore, the pitch command exhibits acceptable variations, ranging between  $15.13^\circ$  and  $15.15^\circ$ , which highlights the controller's ability to maintain stable pitch control compared to other controllers. These favorable outcomes

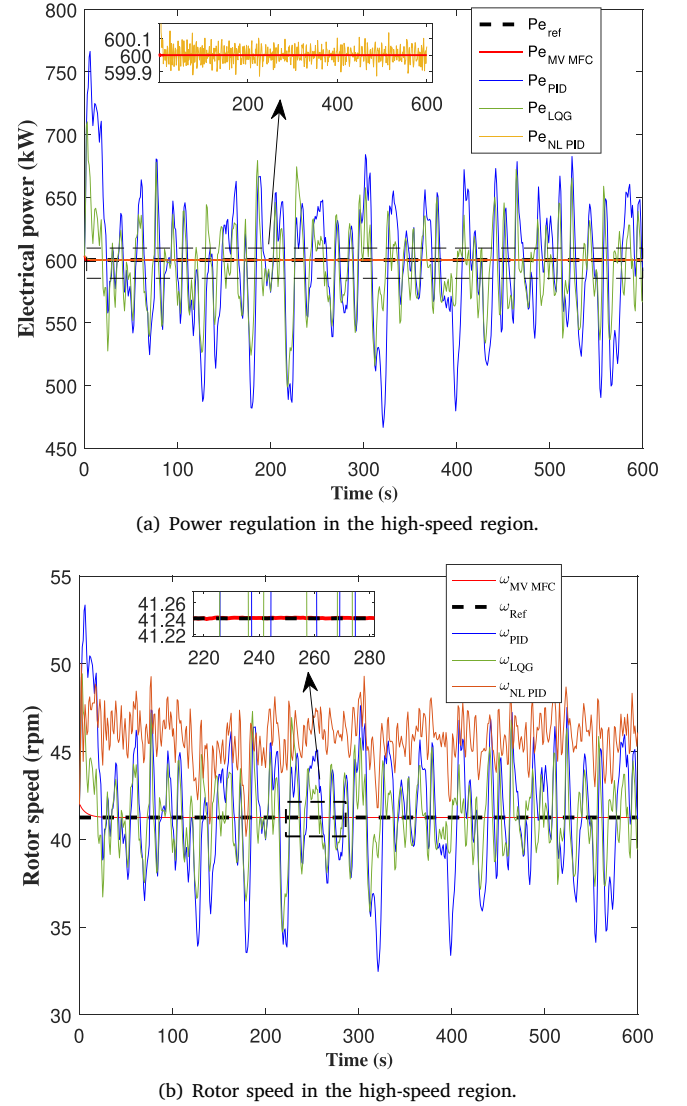


Fig. 20. Comparison between different controllers in the high-speed region.

demonstrate the effectiveness and reliability of the model-free multivariable control approach in ensuring precise and stable operation for the wind turbine system.

The proposed technique, incorporating both mathematical models and the FAST simulator, exhibits notable advancements over classical control approaches, namely Non-linear, PID, and LQG. Two key aspects stand out from this comprehensive analysis. Firstly, the technique achieves exceptional precision in regulating electrical power and rotor speed, ensuring strict adherence to operational norms while maintaining minimal standard deviations. This precision significantly outperforms traditional control methodologies. Secondly, the technique guarantees stable control torque signals, operating at their optimal values with minimal deviation, representing a substantial improvement compared to classical control approaches, which are known for their higher signal variations. The developed MV-MFC approach convincingly demonstrates its efficacy and superiority in controlling power systems. This research sets a promising direction for further advancements in wind turbine control strategies, offering a solid foundation for optimizing the efficiency and performance of renewable energy systems.



## 5. Conclusion

The work discussed model-free controllers design for a variable-speed wind turbine in both operating regions. The simulation findings show that the developed model-free controllers have great tracking performance. Simultaneously, the algebraic estimator proved to be a feasible alternative to conventional state observers, allowing quick and accurate estimation with strong closed-loop performance without the need for a model-based observer. The model-free control outperformed the existing classical control techniques, especially when evaluated using the FAST simulator. As a result, model-free control is a practical and simple alternative for regulating wind turbine systems where obtaining an accurate dynamical model is challenging.

Nevertheless, some limitations should be noted. The performance depends on the accuracy of the real-time algebraic estimation, which can be affected under strong turbulence or noisy measurements. Furthermore, although tuning is simpler compared to model-based controllers, it still requires careful selection of the estimation window and control gains. In the multivariable case, the need for simultaneous estimation of several dynamics slightly increases computational effort. Despite these points, the proposed model-free approaches remain highly attractive for efficient and robust wind turbine control.

## CRedit authorship contribution statement

**Laid Sehili:** Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Boubekeur Boukhezzar:** Writing – review & editing, Validation, Supervision, Resources. **Cedric Join:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis.

## Ethical statement

The corresponding author assures that the following is fulfilled:

(1) The work described has not been published previously except in the form of a preprint, an abstract, a published lecture, academic thesis or registered report.

(2) The paper is not currently being considered for publication elsewhere.

(3) the article's publication is approved by all authors and tacitly or explicitly by the responsible authorities where the work was carried out.

(4) if accepted, the article will not be published elsewhere in the same form, in English or in any other language, including electronically without the written consent of the copyright-holder.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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